PROGRAM-9

1. Problem Statement

Develop a travel budget analysis system that converts travel expenses from different currencies into a selected base currency using supplied exchange rates. The system should include currency conversion fees, calculate total travel expenses, compare planned spending against the allocated budget, and identify the largest expense category. This helps travelers manage their expenses and avoid exceeding their budgets.

2. Objectives

1. **To accept travel expenses in multiple currencies.**
2. **To convert expenses into a base currency using supplied exchange rates.**
3. **To calculate additional currency conversion fees.**
4. **To calculate the total travel expenditure.**
5. **To compare planned spending with the allocated travel budget.**
6. **To identify the largest expense category.**
7. **To determine whether the traveler is within the budget or has exceeded it.**

3. Key Features

- **Multi-Currency Support: Accept expenses in different currencies, such as USD, EUR, GBP, and INR.**
- **Currency Conversion: Convert expenses into a selected base currency using the given exchange rates.**
- **Conversion Fee Calculation: Include additional fees charged during currency conversion.**
- **Expense Categorization: Organize expenses into accommodation, transportation, food, shopping, and entertainment.**
- **Budget Tracking: Compare total expenses against the planned budget.**
- **Largest Expense Identification: Identify the category that accounts for the highest expenditure.**
- **Remaining Budget Calculation: Display the amount remaining after expenses.**
- **Budget Status Report: Indicate whether the traveler is within budget or over budget.**

4. Applications

1. **Personal travel planning.**
2. **International tourism.**
3. **Business trips and corporate travel.**
4. **Student educational tours.**
5. **Travel agencies and tour operators.**
6. **Family vacation budget management.**
7. **International business expense tracking.**

5. Example Calculation

Suppose a traveler plans a trip with a budget of ₹50,000.

Travel Expenses

Expense Category	Original Amount	Currency	Exchange Rate	Amount in INR
Accommodation	**200**	**USD**	**₹85**	**₹17,000**
Transportation	**100**	**USD**	**₹85**	**₹8,500**
Food	**80**	**EUR**	**₹95**	**₹7,600**

Expense Category	Original Amount	Currency	Exchange Rate	Amount in INR
Shopping	**50**	**EUR**	**₹95**	**₹4,750**
Entertainment	**30**	**USD**	**₹85**	**₹2,550**
Total				**₹40,400**

Note: The exchange rates are hypothetical and are provided only for this example.

Conversion Fee

Assume the currency conversion fee is 2% of the converted expense total.

Conversion Fee = ₹40,400 × 2%

Conversion Fee = ₹808

Total Travel Expenditure

Total Expenditure = Converted Expenses + Conversion Fee

Total Expenditure = ₹40,400 + ₹808

Total Expenditure = ₹41,208

Budget Analysis

Description	Amount
Planned Budget	**₹50,000**
Total Expenditure	**₹41,208**
Remaining Budget	**₹8,792**
Budget Status	**Within Budget**

Largest Expense Category

Accommodation costs ₹17,000, making it the largest expense category.

Result: The traveler spends ₹41,208, remains within the ₹50,000 budget, and has ₹8,792 remaining. Accommodation is the largest expense category.

6. Technologies Used

- **Programming Language: Python**
- **User Interface: HTML, CSS, and JavaScript (optional)**
- **Database: MySQL or SQLite (optional)**

7. Expected Outcome

The system will provide an organized analysis of travel expenses in multiple currencies. It will calculate conversion fees, track total spending, identify the largest expense category, and determine whether the travel budget has been exceeded. This will help travelers make informed financial decisions and manage their travel expenses effectively.

Programm

```
# Multi-Currency Travel Budget Analysis


# Input budget and conversion details

budget = float(input("Enter travel budget in INR: "))

exchange_rate = float(input("Enter exchange rate (1 foreign currency unit in INR): "))

fee_percentage = float(input("Enter currency conversion fee percentage: "))
```

```
# Input travel expenses
n = int(input("Enter number of expense categories: "))

expenses = {}
total_expenses = 0
total_fees = 0

for i in range(n):
    print("\nExpense Category", i + 1)

    category = input("Enter category name: ")
    amount = float(input("Enter expense amount in foreign currency: "))

    # Convert expense into INR
    converted_amount = amount * exchange_rate

    # Calculate conversion fee
    conversion_fee = converted_amount * fee_percentage / 100

    # Total cost including conversion fee
    total_cost = converted_amount + conversion_fee

    expenses[category] = total_cost
    total_expenses += converted_amount
    total_fees += conversion_fee

    print("Converted Amount: Rs.", round(converted_amount, 2))
    print("Conversion Fee: Rs.", round(conversion_fee, 2))
    print("Total Cost: Rs.", round(total_cost, 2))

# Calculate final spending
```

```
total_spending = total_expenses + total_fees
remaining_budget = budget - total_spending

# Identify largest expense category
if expenses:
    largest_category = max(expenses, key=expenses.get)
    largest_amount = expenses[largest_category]
else:
    largest_category = "None"
    largest_amount = 0

# Display travel budget report
print("\n======= TRAVEL BUDGET REPORT =======")
print("Travel Budget: Rs.", round(budget, 2))
print("Total Expenses: Rs.", round(total_expenses, 2))
print("Total Conversion Fees: Rs.", round(total_fees, 2))
print("Total Spending: Rs.", round(total_spending, 2))
print("Largest Expense Category:", largest_category)
print("Largest Category Cost: Rs.", round(largest_amount, 2))

if remaining_budget >= 0:
    print("Remaining Budget: Rs.", round(remaining_budget, 2))
    print("Status: Within Budget")
else:
    print("Budget Exceeded By: Rs.", round(abs(remaining_budget), 2))
    print("Status: Over Budget")

print("====================================")
```

output:

```
C:\Users\hp\PycharmProjects\PythonProject48\.venv\Scripts\python.exe
C:\Users\hp\PycharmProjects\PythonProject48\problem-9.py
Enter travel budget in INR: 50000
```

Enter exchange rate (1 foreign currency unit in INR): 20000

Enter currency conversion fee percentage: 50

Enter number of expense categories: 65

Expense Category 1

Enter category name: vijayawada travels

Enter expense amount in foreign currency: 789

Converted Amount: Rs. 15780000.0

Conversion Fee: Rs. 7890000.0

Total Cost: Rs. 23670000.0

Expense Category 2

Enter category name: guntur travels

Enter expense amount in foreign currency: 564